Rishi Rakesh Agrawal

Junior Year Undergraduate, Department of Mechanical Engineering, IIT Kanpur
rishira21@iitk.ac.in | +91-7861061362
Rishi Agrawal | Rishi1704 | Rishi

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech in Mechanical Engineering	Indian Institute of Technology Kanpur	8.44/10
2021	Class XII (GSEB)	Green Valley Higher Secondary School	91.38%
2019	Class X (GSEB)	Gujarat Public School	84.84%

Scholastic Achievements

- Received **OPJEMS Award** in academic year **2022-23** for being in top 100 students across 45+ Premier Institutes
- Awarded **Academic Excellence Award** twice in academic years **2021** and **2022** for exceptional performance in academics
- Achieved **AIR 2566** in **JEE Advanced 2021** and **AIR 2980** in **JEE Mains 2021** among 1.2+ million candidates

Work Experience

Student Research Associate || **Industry 4.0** | *May'23 - Present*

Project Supervisor: Prof. Nalinaksh S. Vyas, Dept. of Mechanical Engineering, IIT Kanpur

Objective	• Development of Indigenous **Industry 4.0** Software and Hardware Platform for Ordnance Production
Approach	• Implemented & Integrated **Enterprise Resource Planning** System to automate **350+** machine operations • Installed cutting edge **Radio Frequency Identification** on **15+** machines enabling inventory management • Implemented **Real Time-Machine Monitoring** by obtaining data through **NI-OPC** servers on **LabView**
Impact	• Improved overall **Productivity** by **17%**, **Diagnosis**, **maintainability** along **Dynamic** process planning

Full Stack Unity Game Developer || *Vizuara* | *Apr'23 - Jun'23*

Objective	• Development of high quality **interactive** digital textbooks at the intersection of Education and **Gamification**
Approach	• Overseen end-to-end game development process, encompassing **Programming**, **3D Art**, **UI/UX**, **Quality**
Impact	• **Elevated educational experience** by delivering cutting edge 3D, gamified & interactive contents to schools

Key Projects

Course Project: Wax Ball Making Machine || *Prof. Mohit Law, Dept. of Mechanical Engineering* *Jan'23 - May'23*

Objective	• Implemented theoretical knowledge to design an **efficient** & **cost-effective** solution for producing wax balls
Approach	• Utilized **Engineering principles** and **mathematical analysis** to accurately calculate torque specifications • Incorporated design considerations for utilization of **1 N.m motor**, **ease of assembly** and **maintenance** • Employed various fabrication techniques, including **lathing**, **milling**, **3D printing** & **CNC** for assembly
Impact	• **Revolutionized** the Wax Ball industry by introducing **automation** resulting in **95%** reduction in defects

Project Mentor: Game Anti-Cheat Client || *Game Development Club, IIT Kanpur* *May'23 - Jul'23*

Mentor	• Mentored **10 freshers** by guiding on fundamentals of **Anti-Cheat**, led development of **Cheat Detection**
Approach	• Researched on various hacking techniques, including **AimBots** and **WallHacks** and existing hacking software • **Integrated** Anti-Cheat Client ensuring **compatibility**, **performance optimization** & effective anti-hack

Option and Strategies to Hedge a Corpus || *Finance and Analytics Club, IIT Kanpur* *May'22 - Jul'22*

Objective	• Evaluate **16+ Option Strategies** to efficiently **Hedge a Corpus** against market risk and fluctuations
Approach	• Calculated & analysed **Greek Function** & conducted comprehensive **Risk Assessment** for 4 option strategy • Utilized **Opstra** to implement **backtests**, running multiple simulation to evaluate strategy performance

Positions of Responsibility

Coordinator, Game Development Club IIT Kanpur *Apr'23 - Present*

Leadership	• Spearheaded a **3-tier team** of **30+** members to organise events, workshops, sessions for campus community • Coordinated with Core Team to organize Semester Projects, for **100+ students** on topics Anti-Cheat & AI
Initiatives	• Delivered **influential** workshops on GameDev, attracting **100+** motivated developers enhancing their skills
Impact	• Received a **funding** of **INR 1 Lakh+** from **Vizuara** as a result of successful **Industrial Collaboration**

Web Organizer, Udghosh IIT Kanpur *Jul'22 - Oct'22*

Teamwork	• Collaborated with a team of **10+** to gather requirements, design and develop the official website of Udghosh
Impact	• **Strategically** elevated festival budget by **INR 45 Lakh** though website by attracting sponsorship & funding

Relevant Technical Skills

Programming: C++, C, Python, Java **Softwares**: Autodesk Inventor, Fusion 360, Arduino IDE, MySQL, MS Office

Relevant Coursework

Thermodynamics	Mechanics of Solid	Engineering Graphics and Design
Manufacturing Process	Fluid Dynamics	Natures and Properties of Materials
Dynamics	Introduction to Critical Thinking	Introduction to Economics

Extra Curricular Activities

Technical	• Achieved **Rank 1** in **SnT Code**, a 10+ hour Inter Hall Competition demonstrating Problem Solving skills • Secured **Rank 1** representing my Hall in **Takneek'22**, the prestigious Inter Hall Technical Tournament • Participated in Problem Statement of The **IGDC GameDev Challenge** in **Inter IIT Tech Meet 11.0**
Sports	• Secured **Gold** representing IIT Kanpur **Hockey Team** in **Udghosh'22**, Annual Sports Festival of IITK • Participated in **Inferno'22**, the Inter Hall Sports Tournament representing my Hall on **Hockey ground**